

Expansion of Aggregation Dynamics with Master Equations

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Outline

1 Purpose

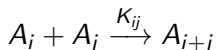
2 Setup

Purpose

- Key Question: How do we avoid gelation in aggregation dynamics with the *least costly* regulatory interference?
- Imagine corporate consolidation, or anywhere else where things tend to get big and we don't want them to. How do we prevent an entity from getting *overwhelmingly* big, ala the giant component, with the *least* possible expenditure of effort?
- Cost: **number of interventions, size of interventions**

Aggregation

Aggregation is generally the transition:



Yields master equation:

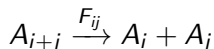
$$\begin{aligned}\frac{dc_k}{dt} &= \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{i \geq 1} K_{ik} c_i \\ &= \frac{1}{2} \sum_{i,j \leq 1} K_{ij} c_i c_j \left[\delta_{i+j,k} - \delta_{i,k} - \delta_{j,k} \right]\end{aligned}$$

Definitions:

- A is a cluster
- i and j are masses
- K_{ij} is the rate of aggregation between clusters of mass i and mass j into a clusters of mass $i+j$
- K , then, is an infinite symmetric matrix $K_{ij} = K_{ji}$
- δ is the Kronecker Delta

Fragmentation

Fragmentation is generally the transition:



Yields master equation:

$$\frac{dc_k}{dt} = \sum_{j \geq 1} F_{kj} c_{j+k} - \frac{1}{2} \sum_{i+j=k} F_{ij}$$

Definitions:

- A is a cluster
- i and j are masses
- F_{ij} is the rate of fragmentation of a cluster of size $i + j$ into a cluster of size i and another of size j

Combined Master Equations

Put together, we have the combined master equation:

$$\frac{dc_k}{dt} = \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{j \geq 1} K_{jk} c_j + \sum_{j \geq 1} F_{kj} c_{j+k} - \frac{1}{2} c_k \sum_{i+j=k} F_{ij}$$

With a competition between the aggregation kernel K and the fragmentation kernel F .

We can now be more specific about our two definitions of the 'cost' of fragmentation. Let l_i , the cost per intervention, be the number of times that the aggregation term F applies. Let l_s instead be the sum of $\min(i, j)$ over all the timesteps F is applied.

Choosing Aggregation Kernel

$$\frac{dc_k}{dt} = \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{j \geq 1} K_{jk} c_j + \sum_{j \geq 1} F_{kj} c_{j+k} - \frac{1}{2} c_k \sum_{i+j=k} F_{ij}$$

Based on polymerization, parameterize aggregation on number of bond points f :

$$K_{ij} = (f - 2)^2 ij + 2(f - 2)(i + j) + 4$$

- Dunbar's number: $f = 152 \implies K_{ij} = \alpha(150^2 ij + 300(i + j) + 4)$
- Proposed *homo corporatum* number: $f \rightarrow \infty \implies K_{ij} \approx \alpha(ij)$
- Note: $K_{ij} = ij$ is the same as percolation // erdos-renyi graph.

Choosing Fragmentation Kernel

$$\frac{dc_k}{dt} = \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{j \geq 1} K_{jk} c_j + \sum_{j \geq 1} F_{kj} c_{j+k} - \frac{1}{2} c_k \sum_{i+j=k} F_{ij}$$

Parameterized by γ , a few different models:

Name	Equation
Chipping Model	$F_{ij} = \gamma_0 + \gamma_1(\delta_{i,1} + \delta_{j,1})$
Sum Size	$F_{ij} = \gamma(i + j)$
Product Splitting	$F_{ij} = \gamma(ij)$
Size Based Chipping	$F_{ij} = \gamma(j\delta_{i,1} + i\delta_{j,1})$

Connect back to goal: how well does each prevent gelation? How much does the intervention of each cost?